

THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

Valerio Dionisi

June 2026

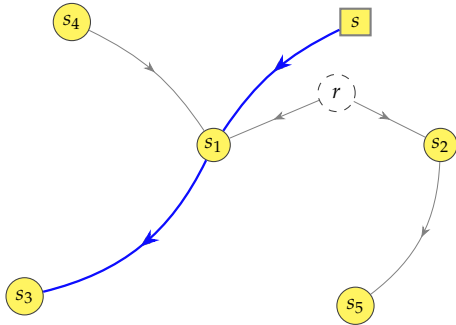
Department of Economics, Management and Statistics

University of Milano-Bicocca

BACKGROUND

- ▶ Comovement across sectors is an essential feature of business cycles {e.g., Lucas (1977)}
 - driven by independent micro-level shocks, not aggregate {e.g., Gabaix (2011), Foerster *et al.* (2011)}
- ▶ Production networks $\longrightarrow$ role of Input-Output linkages {e.g., Long and Plosser (1983)}
 - an idiosyncratic shock propagates to others along supply chains $\leftrightarrow$ “cascade” effects;
 - economically dominant sectors generate aggregate fluctuations {e.g., Acemoglu *et al.* (2012)}.
- Channel: *vertical complementarities*, through network trade intensities {e.g., Shea (2002)}
 - *linear economies* $\leftrightarrow$ comovement from even propagation of shocks {most of the literature};
 - *non-linear complementarities in production* $\leftrightarrow$ the even transmission of shocks is reversed, thereby allowing opposite comovement {e.g., Corsetti *et al.* (2008), Baqaee and Farhi (2019)}.

Verticality



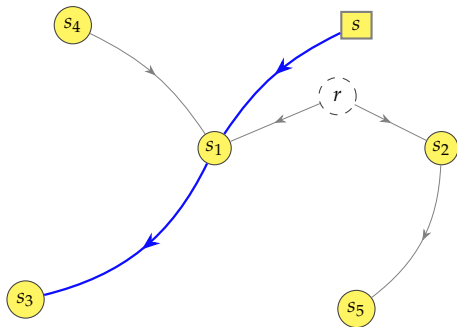
THIS PAPER: CENTRAL INTUITION

- ▶ Beside the *vertical* dimension, there exists an *horizontal* geometry of the network
 - 👉 sectors are additionally held together by *demand and supply interdependencies from having in common a similar Input-Output structure*
 - 📌 coexistence in the same primitive established by the **REPRESENTATION THEOREM** ▶ Theorem 1
- ▶ Horizontality $\longrightarrow$ network “economic” distances between paired sectors
 - *factor input demand* distance, when *buying from* similar upstream sectors;
 - *factor input supply* distance, when *selling to* similar downstream sectors.
- ▶ These novel horizontal interdependencies add to the vertical transmission of shocks
 - network connections extend beyond trade flows $\leftrightarrow$ horizontal mapping

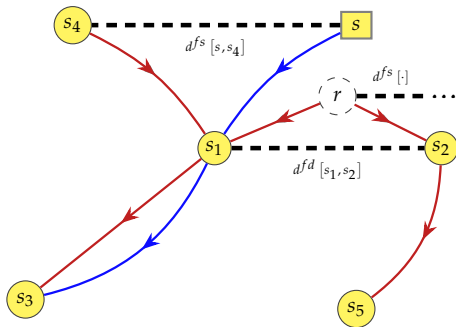


“who trades with whom” + “who depends on whom”

Verticality



Verticality + Horizontality



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NEWS FEED

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by TechNode Feed Jan 22, 2026

Credit: BOE

Goertek Plunges as Chinese Apple Supplier Loses AirPods Pro Order to Luxshare

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► Supply: Micron

► Supply: AirPods

► Supply: OLED panels

► Demand: memory chips

► Demand: cloud

► Demand: TSMC

CENTRAL THEORETICAL RESULTS

Question

how does the horizontal transmission impact comovement and aggregate dynamics?

- ① *sectoral comovement*: it depends on the nature of the network “economic” distance
 - ambiguous in *demand*: production shifts away from common upstream sellers;
 - opposite in *supply*: relative price effect from competition for common downstream buyers;
 - minor distance $\leftrightarrow$ stronger horizontal complementarities across nearby sectors.
- ② *aggregate dynamics*: systematic comovement prevent shocks from averaging out
 - sectors sharing upstream suppliers and downstream buyers respond simultaneously;
 - not through vertical transmission or cascading amplification;
 - break of the “standard diversification argument” by Lucas (1977).

EMPIRICAL OUTCOMES

- ▶ A data-driven simulation validates the aggregate fluctuations results
 - *horizontal* propagation systems on aggregate GDP responses;
 - larger aggregate volatility, slower rate of decay.
- ▶ Sectoral employment comovement is well documented {e.g., Cooper and Haltiwanger (1990), Christiano and Fitzgerald (1998)}, but no studies exploit a networked dimension
- ▶ Empirics on U.S. data ratify the theoretical insights on employment comovement
 - ① demand-linked sectors: *muted/mixed*;
 - ② closely supply-linked sectors: *opposite*; if farther apart: *positive*;
 - ③ clear responses for peripheral sectors after a shock in central sectors: *opposite* and *positive*.

RELATED LITERATURE

👉 Advance the understanding of how Input-Output economies work

- ▶ production networks {e.g., Long and Plosser (1983), Acemoglu *et al.* (2012), Carvalho (2014), Grassi (2017), Oberfield (2018), Carvalho and Grassi (2019), Liu (2019), Baqaee and Farhi (2019), Acemoglu and Azar (2020), vom Lehn and Winberry (2022), La'O and Tahbaz-Salehi (2022), Ghassibe (2024), Huo *et al.* (2025), Qiu *et al.* (2026)}
 - wide range of topics (fiscal and monetary policy, efficiencies and inefficiencies, growth, financial frictions, international contexts), where the network transmission is *vertical* only
 - **Contribution:** introduce, explore, and formalize the network's *horizontal geometry*
- ▶ empirics on the functioning of Input-Output linkages {e.g., Conley and Dupor (2003), Acemoglu *et al.* (2015), Barrott and Sauvagnat (2016), Ghassibe (2021), Barattieri and Cacciatore (2023), Barattieri *et al.* (2023)}
 - **Contribution:** provide evidence of shocks' transmission due to network distances
- ▶ empirics on comovement of sectoral employment {e.g., Cooper and Haltiwanger (1990), Christiano and Fitzgerald (1998), Cassou and Vazquez (1990), Yedid-Levi (2016), Kim (2020)}
 - **Contribution:** take a production network perspective

WHAT'S TO COME

- ① FOUNDATIONAL THEORY, to define network distances and their role on comovement
 - inspect how shocks propagate considering horizontal linkages and input substitutability
- ② AGGREGATE DYNAMICS, to induce aggregate cycles from horizontal complementarities
 - different and independent mechanisms from standard vertical propagation
- ③ EMPIRICS, network effects of sectoral variations to study theoretical results
 - data-driven simulation of aggregate GDP fluctuations;
 - network dimension of nearby sectors' employment changes on sectoral employment.
- ④ POLICY IMPLICATIONS, for future novel perspectives (not today, but in the paper)
 - improving the design of policy interventions through the network's horizontal geometry

FOUNDATIONAL THEORY



GENERAL FRAMEWORK

- ▶ Consider a general production function for sector $\{s, s', s'', \dots, m\} \in \Phi$

$$y(s) = z(s) f^y \left(n(s) ; \{x(s, s')\}_{s' \in \Phi} \right)$$

- ▶ Combination of optimality conditions delivers

$$x(s, s') = f \left(\alpha(s, s'), p(s') ; \alpha(s), n(s), w(s) \right) , \quad \forall s' \in \Phi \quad (1)$$

- $\alpha(s, s')$: intensity of the input from sector- s' in the intermediates bundle of sector- s
- ▶ Then, consider two types of *production network “economic” distances*:
 - ① *factor input demand*, when given sectors are *buying from* the same sector;
 - ② *factor input supply*, when given sectors are *selling to* the same sector.

DISTANCES

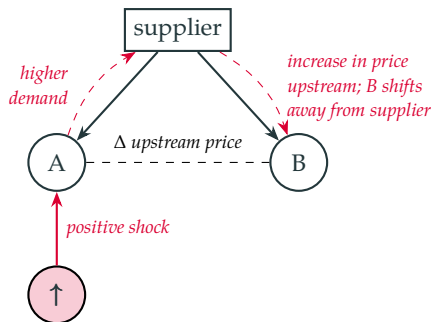
- ▶ Exploit the distance's definitions to relate sectoral optimality conditions ▶ Mixed conditions
 - on the input side, sectors buying from the same suppliers rebalance simultaneously their combination of labour and intermediates; ▶ Demand
 - on the output side, sectors selling to the same buyers exhibit relative price changes, and thus variations in downstream demand. ▶ Supply
- ▶ Systematic adjustments scaled by *distance matrices*, $\mathcal{D}^j$ for $j = \{fd, fs\}$ ▶ Interpretation

 **THEOREM 2** formalizes their characteristics, such as: ▶ Theorem 2

- *non-negative* and *full*: all sectors are horizontally connected, even if not trading vertically;
- a unique Input-Output structure has different network distance relations;
- conditions under which distances are uniquely determined (demand = supply). ▶ Corollary 1

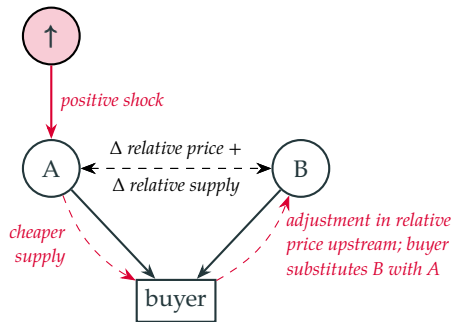
GRAPHING THE ECONOMICS OF DISTANCE-LEAD COMOVEMENT

Demand distance: upstream pass-through



downstream shifting away from upstream input
 $\Rightarrow$ vertical price effect + horizontal ambiguity downstream

Supply distance: downstream pass-through



upstream inputs used jointly downstream
 $\Rightarrow$ demand reallocation and clear horizontal transmission

AGGREGATE DYNAMICS

TO WHAT EXTENT HORIZONTALITY CONTRIBUTES TO AGGREGATE DYNAMICS?

 **THEOREM 5** → network distances alone generate persistent aggregate fluctuations

$$\liminf_{m \rightarrow \infty} \text{var}(Y_m) > 0$$

- even absent vertical production chains;
- they ensure persistence, while granularity magnifies aggregate volatility;
- horizontal propagation breaks the Lucas (1977)'s diversification argument.

► Theorem 5

► Corollary 3

► Corollary 4

- While *vertical propagation* inherently amplifies idiosyncratic shocks ...
 - as shocks feed into sectors along production chains {e.g., Long and Plosser (1983)}
 - thereby generating cascading effects and aggregate responses {e.g., Acemoglu et al. (2012)}
- ... *horizontal propagation* induces comovement and not recursive feedback loops
 - sectors sharing suppliers or buyers respond simultaneously to horizontal shocks
 - no cascades, but systematic correlation that survives aggregation

VERTICAL AND HORIZONTAL NON-DEPENDENCE

- ▶ Finally, a dimensional separation is established

 **THEOREM 3:** the horizontal geometry remains unweighted by vertical intensities

- the distance matrix is not merely a transformation of the Input-Output matrix, but a complementary object that reveals otherwise hidden propagation channels

▶ Theorem 3

 **THEOREM 4:** the horizontal geometry is increasingly important as the network expands

$$\left\| \mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m \right\|_2 / \left\| \mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m \right\|_2 \xrightarrow{m \rightarrow \infty} 0$$

- for larger production networks (e.g., firm-to-firm network), vertical buyer-supplier chains become too sparse, while horizontal complementarities grow denser;
- even when vertical chains expand endogenously as the network grows in size.

▶ Theorem 4

▶ Corollary 2

FIX IDEAS: MAIN THEORETICAL TAKEAWAYS

- ▶ Horizontal transmission is about a rebalancing across factors of production ▶ Empirics
 - *endogenous comovement*: changes in intermediate inputs induce analogous labour changes
- 👉 Opposite comovement for short distances, positive comovement for longer distances
 - potentially *ambiguous* under demand distances, *clear* under supply distances ...
 - ... but connecting all network participants, even those disconnected;
 - substitution elasticities matter in magnitude, not in the direction $\{\neq \text{Baqae and Farhi (2019)}\}$.
- 👉 Aggregate fluctuations induced by horizontal complementarities alone
 - shocks propagate along common buyers and suppliers, inducing mutual responses ...
 - ... no amplification through cascades but network-wide correlations;
 - breaking of the “standard diversification argument” → higher aggregate volatility.

EMPIRICS



- ▶ From the United States' Bureau of Economic Analysis (BEA) I collect information, for 3-digit U.S. 2017 NAICS sectors on:

▶ I-O data

▶ I-O structure

- inter-sectoral linkages exploiting Input-Output (I-O) matrices:
 - ★ share of commodity production (*market share, after redefinitions* tables);
 - ★ I-O weights in producer values (*commodity-by-commodity total direct requirements* tables).
 - disregard small I-O transactions for vertical effects {e.g., [Acemoglu et al. \(2012\)](#), [Carvalho \(2014\)](#)}
 - ★ no link if sector-s's total input sales/purchases are less than 1%
 - employment, value-added, imports and exports, aggregate TFP, public expenditure.
- ▶ *Unit*: 65 private sectors at 3-digit U.S. 2017 NAICS
 - ▶ *Period*: 1998-2022
 - ▶ *Network structure* at 2007

robust at different years

SETTING THE STAGE

- Degree of separation between sectors within the network (low value, low distance)

- entries of distance matrices are *Euclidean distances*, so that $d^j[s, s'] \in [0, 1]$ ► Intensive margin

$$d^{fd}[s, s'] = \sqrt{[\alpha(s, s) - \alpha(s', s)]^2 + [\alpha(s, s') - \alpha(s', s')]^2 + \dots + [\alpha(s, m) - \alpha(s', m)]^2}$$

from the *directed* Input-Output network, and not its Leontief transformation

- by Theorem 3, consider *extensive margins*, $d(s, s') = \{1, 2, \dots, d_{max}\}$ ► Graph ► Characteristics

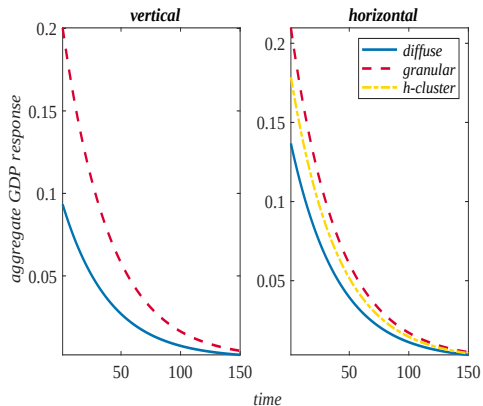
★ computed as a *shortest path problem* (how many steps to reach other nodes) ► Algorithm

- Before the empirics, I simulate aggregate GDP responses to sectoral AR(1) shocks

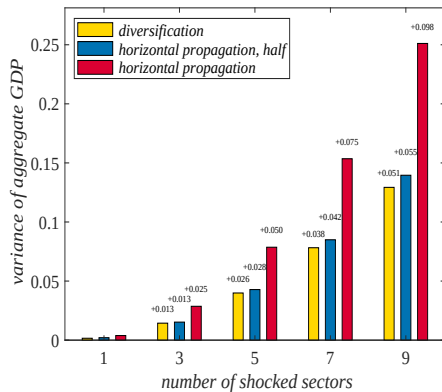
$$Y_t = \boldsymbol{\lambda}' \mathbf{y}_t \quad \longleftrightarrow \quad Y_t = \boldsymbol{\lambda}' \mathcal{P}^{(\mathcal{V}, \mathcal{D})} (\rho_t \boldsymbol{\epsilon}_0)$$

exploiting the network's vertical (Input-Output chains) and horizontal (shared buyers/suppliers) propagation structures, relying on observed matrices and data

$$\|\mathcal{P}^{(\mathcal{D})}\|_2 = 0.156 > 0.124 = \|\mathcal{P}^{(Lucas, 1977)}\|_2$$



(a) Response



(b) Volatility

Figure 3: Aggregate fluctuations

PROPAGATING PATTERNS

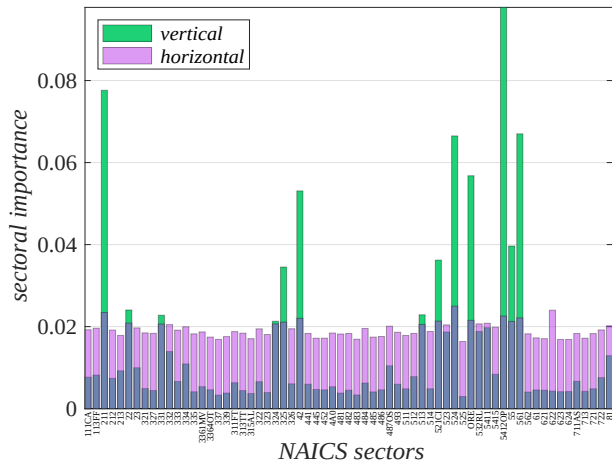


Figure 4: Sectoral influence in shock transmission

VALIDATING HORIZONTAL GEOMETRIES: A BARTIK-STYLE DESIGN

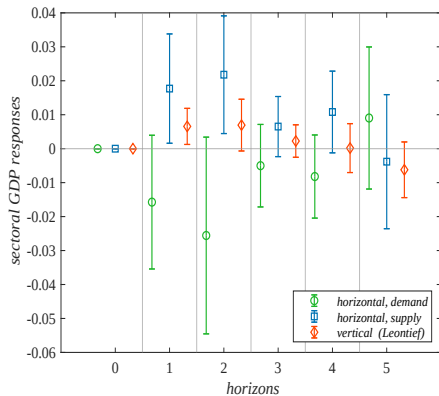
- ▶ So far, from sectoral to aggregate $\leftrightarrow$ now, aggregate shocks hit all simultaneously
 - *cross-sectional variation*: sectors differ in their *network exposure* to the same disturbance G_t
- ▶ Network structure interacting with predetermined sensitivities to aggregate shock
 - horizontal exposure *residualised* from orthogonal vertical (Leontief inverse) exposure, $\hat{r}^{\mathcal{D}^j}$

- ▶ *Shift-Share Local Projection* (LP-IV) estimation

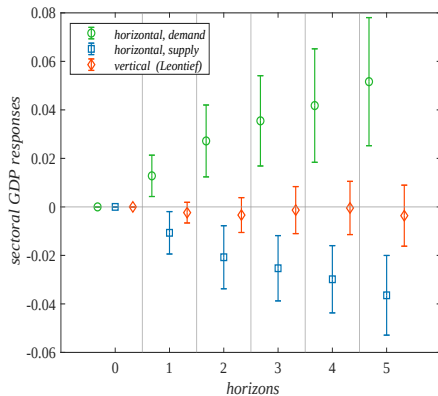
$$\begin{aligned}\Delta y_{t+h}(s) &= \beta_h^{\mathcal{D}^{fd}} \left(G_t \times \hat{r}^{\mathcal{D}^{fd}}(s) \right) + \beta_h^{\mathcal{D}^{fs}} \left(G_t \times \hat{r}^{\mathcal{D}^{fs}}(s) \right) \\ &\quad + \beta_h^{\mathcal{V}} \left(G_t \times exposure^{\mathcal{V}}(s) \right) + \phi(t+h) + \phi_h(s) + v_{t+h}^y(s)\end{aligned}$$

- $\beta_h^{\mathcal{D}^j}$ measures *differential* sectoral responses across exposure levels, not unconditional
- ▶ Identification is purely cross-sectional
 - at each horizon- h , if $\beta_h^{\mathcal{D}^j} > 0 \Rightarrow$ amplification; if $\beta_h^{\mathcal{D}^j} < 0 \Rightarrow$ attenuation via the network

NETWORK EXPOSURES AND AGGREGATE SHOCKS



(a) Total Factor Productivity



(b) Government spending

Figure 5: “Networking up” aggregate shocks

AN OVERVIEW OF THE EMPIRICAL RESULTS ON SECTORAL COMOVEMENT

- ▶ The results thus far focused on the “mechanics”, not on sectoral comovement
- ▶ To study comovement, I follow the framework of [Barattieri and Cacciatore \(2023\)](#)
 - IDENTIFICATION, to estimate sectoral shocks by extracting exogenous dynamics;
 - LOCAL PROJECTIONS, use the estimated shocks for short-run dynamic responses.
- ① Increasing the set of intermediate inputs leads to a smaller rise in its employment
 - minimal but alternate effects from similar changes in distanced sectors {Figures [D.1-D.4b](#)}
- ② Sectoral comovement of employment {Figures [E.1-E.2](#); Figures [E.3-E.4](#)}
 - closer supply distance → opposite employment comovement;
 - closer demand distance → ambiguous responses, yet dampening the vertical channel;
 - positive comovement emerges among more distantly demand- and supply-related sectors.
- ③ Results on employment comovement hold along different robustness checks ▶ Robustness
 - peripheral sectors, changes in sectoral inventories, different base year for the network

POLICY IMPLICATIONS

- 👉 Horizontal complementarities are crucial for the transmission of sectoral variations
- ▶ *Fiscal policy*, typically generating upstream transmission { e.g., [Barattieri et al. \(2025\)](#) };
 - it works by expanding sector-specific demand, and thus the supply of others;
 - horizontal geometry is at the core of the effects from government spending;
 - targeted government spending, fiscal multiplier, tariff policy?
- ▶ *Industrial policy*, characterised by two crucial results:
 - market imperfections accumulate through backward demand linkages { [Liu \(2019\)](#) }, and positive effects on buyers of targeted sectors { [Lane \(2025\)](#) };
 - how do interventions affect untargeted sectors exhibiting similar Input-Output structure?
- ▶ *Transmission mechanism*, role of (consumption or production) complementarities
 - asymmetric supply shock is a demand-like shock at the aggregate level { [Guerrieri et al. \(2022\)](#) };
 - horizontal complementarities may clearly, easily and linearly identify demand and supply forces within the network that guide such Keynesian transmission.

- ▶ On the modelling side, expanded framework for:
 - global production networks, extending beyond vertical supply chains {e.g., [Huo et al. \(2025\)](#)};
 - firm-to-firm networks, showing high sparsity {e.g., [Bohem et al. \(2019\)](#), [Carvalho and Grassi \(2019\)](#)};
 - endogenous network formation and inputs specificity {e.g., [Carvalho and Voigtländer \(2015\)](#)};
 - complementarity and sectoral shocks to aggregate outcomes {e.g., [Atalay \(2017\)](#)}.
- ▶ Horizontal geometry is well suited for broader class of economic interdependencies, including investment {e.g., [vom Lehn and Winberry \(2022\)](#)}, credit {e.g., [Delli Gatti et al. \(2010\)](#)}, and financial {e.g., [Acemoglu et al. \(2015\)](#), [Huremovic et al. \(2025\)](#)} networks

CONCLUSION

TAKE-AWAY REMARKS

- ✱ *Horizontal dimension* of a network complements the vertical dimension
 - 👉 it is not the mere existence of Input-Output connections, but **their demand-supply geometry that critically shapes the transmission of micro-originated shocks**
- ▶ Horizontal geometry captured by *network “economic” distances* between sectors
 - ① demand-based distances work through production inputs dynamics, while supply-based distances induce opposite comovement due to relative prices and revenue reallocation;
 - ② persistent aggregate effects even without vertical cascading effects and central sectors;
 - powerful and independent refinement to prevailing network analysis.
- ▶ Comprehensive framework to build distance matrices and form theoretical insights
 - test and corroborate theory-based predictions on sectoral U.S. Input-Output data;
 - suitable for analysis involving the network structure and to design policy prescriptions.

APPENDIX

- ▶ February 2, 2026 ↔ Samsung Electronics becomes more technically sophisticated in producing more performative HBM4 memory chips
 - ▶ [Barron](#) ▶ [investing.com](#) ▶ [MarketWatch](#)
 - anticipated production for the Nvidia's AI processor, buying also from Micron;
 - Micron's stock price declines despite solid fundamentals, while Samsung's increases.
- ▶ Network-based mechanism through *common downstream buyer*
 - Samsung and Micron → slightly different chips but same customer, Nvidia;
 - positive supply shock to Samsung → earlier and better production;
 - investors expect lower margins for Micron → future downstream demand reallocation.
- 👉 Horizontal transmission through Input-Output proximity
 - *supply-based* propagation of an idiosyncratic shock without direct network linkages

- ▶ *November 2022* ↔ Goertek loses major Apple AirPods Pro 2 assembly contract, which is reassigned to Luxshare Precision
 - ▶ [Yicai Global](#) ▶ [Euronews](#) ▶ [MacRumors](#)
 - Apple reallocates orders due to production/quality issues at Goertek for AirPods Pro 2;
 - Luxshare gains additional orders and revenue from the same Apple demand, while Goertek experiences a significant revenue reduction (CNY 3.3 billion, USD 455 million).
- ▶ Network-based mechanism through *common downstream buyer*
 - Goertek and Luxshare → both supply components to Apple;
 - negative demand shock in Goertek → Apple reallocates orders to Luxshare;
 - propagation through demand → Luxshare benefits, Goertek loses.
- 👉 Horizontal transmission through Input-Output proximity
 - *supply-based* propagation of an idiosyncratic shock without direct network linkages

- ▶ *January 2026* ↔ Apple reallocates iPhone OLED panel orders from BOE to Samsung Display due to production/quality issues at BOE
 - ▶ [MacRumors](#)
 - ▶ [TechNode](#)
 - ▶ [DisplayDaily](#)
 - BOE faces persistent production/quality issues and Apple cannot meet launch timelines;
 - Apple reallocates millions of OLED panel orders to Samsung Display, which gains additional business, while BOE loses revenues and market share.
- ▶ Network-based mechanism through *common downstream buyer*
 - BOE and Samsung Display → both supply components (OLED panels) to Apple;
 - negative supply shock in BOE → troubles in production efficiency to delivering orders;
 - Apple reallocates its demand → Samsung Display benefits, BOE loses.
- 👉 Horizontal transmission through Input-Output proximity
 - *supply-based* propagation of an idiosyncratic shock without direct network linkages

- ▶ 2024-2027 $\leftrightarrow$ surge in demand for DRAM and HBM memory chips driven by AI and cloud data centres

▶ [investing.com](#)

▶ [Business Standard](#)

▶ [Reuters](#)

- Samsung, SK Hynix, Micron and other memory suppliers prioritize AI and cloud orders;
 - supplier capacity is limited, causing higher prices and longer lead times;
 - other sectors (*e.g.*, automotive, PC, smartphones) sharing the same suppliers experience indirect supply constraints and cost increases.
- ▶ Network-based mechanism through *common upstream seller*
 - AI/cloud workloads and other sectors $\rightarrow$ shared memory suppliers;
 - idiosyncratic demand shock in AI/cloud $\rightarrow$ capacity reallocation at suppliers;
 - increase in upstream price $\rightarrow$ other sectors face indirect constraints and cost pressures.
- 👉 Horizontal transmission through Input-Output proximity
 - *demand-based* propagation of an idiosyncratic shock without direct vertical linkages

- ▶ 2023-2024 $\leftrightarrow$ surge in AI demand increases usage of cloud-based GPU capacity provided by AWS or Microsoft Azure
 - ▶ Times of India
 - ▶ CNBC
 - ▶ Reuters
 - ▶ Financial Times
 - AI users rapidly expand demand for high-end GPUs (e.g., Nvidia H100);
 - cloud providers report capacity constraints and prioritize large AI clients; smaller or non-AI users face higher prices or limited access to GPU instances.
- ▶ Network-based mechanism through *common upstream seller*
 - AI and gaming/SaaS/fintech sectors $\rightarrow$ unrelated activities but same cloud provider;
 - demand shock in AI sector $\rightarrow$ providers face GPU capacity constraints;
 - upstream price adjustments $\rightarrow$ other sectors experience cost and quantity effects.
- 👉 Horizontal transmission through Input-Output proximity
 - *demand-based* propagation of an idiosyncratic shock without direct network linkages

- ▶ *January-February 2026* ↔ surge in AI-chip demand generates unprecedented pressure on TSMC's advanced-node capacity

▶ [MacRumors](#)

▶ [Business Insider](#)

▶ [AppleInsider](#)

- exceptional demand for AI accelerators increases priority allocation at TSMC;
- Apple faces tighter access to cutting-edge production nodes.

- ▶ Network-based mechanism through *common upstream seller*

- AI chip producers and Apple → different final markets but same foundry (TSMC);
- positive demand shock in AI sector → capacity reallocation at TSMC;
- reduced effective supply for Apple → production inputs potentially from others.

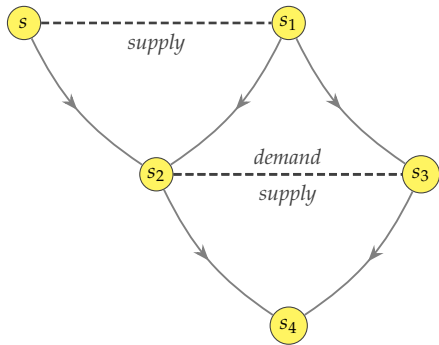
👉 Horizontal transmission through Input-Output proximity

- *demand-based* propagation of an idiosyncratic shock without direct network linkages

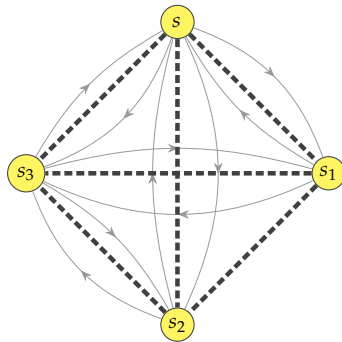
ASSUMPTION 1 (Production technology requirements)

As main characteristics of a production function: (i) it has constant returns to scale in either $x(s, s' \in \Phi)$ and $n(s)$, so that production inputs shares sum to one, $\alpha(s) + \sum_{s'} \alpha(s, s') = 1$; (ii) it is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in $z(s)$, $x(s, s' \in \Phi)$, and $n(s)$; (iii) the case $f^y(0, \cdot)$ is ruled out as labour input is essential to production; (iv) at least two elements of $x(s, s' \in \Phi)$ have to be positive, thus $f^y(\cdot, 0)$ cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector, $x(s, s) > 0$, $\forall s \in \Phi$, that is a production plan with a bundle of intermediate inputs as $f^y(\cdot, x(s, s' \in \Phi_s))$ is precluded.

- note that any functional form chosen for the production function must satisfy the above regularity conditions



(a) Distance definition



(b) Distance network

Figure A.1: Horizontal Input-Output geometries

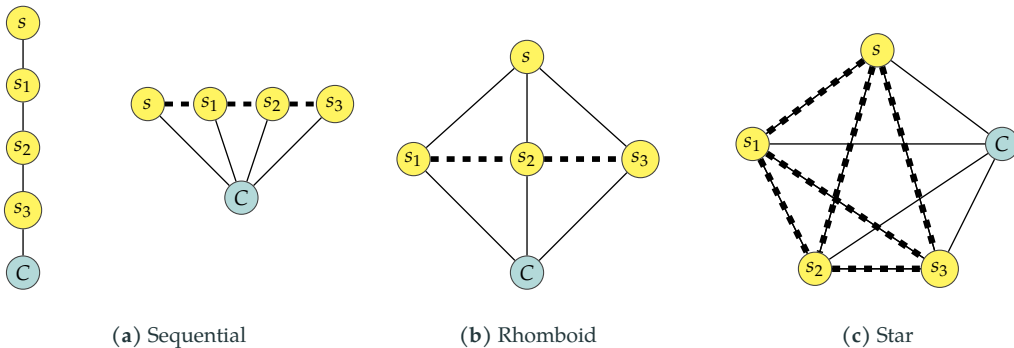


Figure A.2: Benchmark economies

- Solving the system for *factor input demand* network distance

$$\begin{cases} x(s, s^*) = \alpha(s, s^*) w(s) n(s) \alpha(s)^{-1} \frac{1}{p(s^*)} \\ x(s', s^*) = \alpha(s', s^*) w(s') n(s') \alpha(s')^{-1} \frac{1}{p(s^*)} \end{cases}$$

would deliver the condition $\frac{n(s')}{x(s', s^*)} = \frac{\alpha(s, s^*)}{\alpha(s', s^*)} \frac{n(s)}{x(s, s^*)}$.

- Solving the system for *factor input supply* network distance

$$\begin{cases} x(s^*, s) = \alpha(s^*, s) w(s^*) n(s^*) \alpha(s^*)^{-1} \frac{1}{p(s)} \\ x(s^*, s') = \alpha(s^*, s') w(s^*) n(s^*) \alpha(s^*)^{-1} \frac{1}{p(s')} \end{cases}$$

would deliver the condition $\frac{x(s^*, s)}{x(s^*, s')} = \frac{\alpha(s^*, s)}{\alpha(s^*, s')} \frac{p(s')}{p(s)}$.

- ▶ Substituting out eq. (1) for two sectors, say $\{s, s'\}$, actually *buying* their intermediate inputs from the same sector, say s^* , then total *log*-differentiation leads to

$$\underbrace{\left[\gamma_{s'}^B d \log n(s') - \delta_{s'}^B d \log x(s', s^*) \right]}_{\Delta \text{ relative input share of sector-} s'} = \underbrace{\frac{\alpha(s, s^*)}{\alpha(s', s^*)}}_{d_{s^* \rightarrow [s, s']}} \underbrace{\left[\gamma_s^B d \log n(s) - \delta_s^B d \log x(s, s^*) \right]}_{\Delta \text{ relative input share of sector-} s}$$

- ▶ Iteratively, the above condition can be written for all sectors from which the pair is buying. Stacked the outcome for all sectors, then

▶ Lemmas

$$\mathcal{F}_{m \times m} = \mathcal{D}_{m \times m}^{fd} \quad (\text{A.1})$$

- 👉 In other words, the comovement of inputs of production across any pair of sectors is determined by their distance in the production network.

- ▶ Substituting out eq. (1) for two sectors, say $\{s, s'\}$, actually *selling* their intermediate inputs to the same sector, say s^* , then total *log*-differentiation leads to

$$\underbrace{\left[\gamma_{s'}^Q d \log x(s^*, s) - \delta_{s'}^Q d \log x(s^*, s') \right]}_{\Delta \text{ relative quantity of intermediate inputs}} = \underbrace{\frac{\alpha(s^*, s)}{\alpha(s^*, s')}}_{d_{\rightarrow s^*}[s, s']} \underbrace{\left[\gamma_s^P d \log p(s') - \delta_s^P d \log p(s) \right]}_{\Delta \text{ relative price of intermediate inputs}}$$

- ▶ Iteratively, the above condition can be written for all sectors to which the pair is selling. Stacked the outcome for all sectors, then

► Lemmas

$$\mathcal{R}_{m \times m} = \mathcal{D}_{m \times m}^{fs} \quad (\text{A.2})$$

- 👉 In other words, the comovement across (marginal) revenues of production across any pair of sectors is determined by their distance in the production network.

LEMMA 1 (Factor input demand)

Changes in the ratio of production inputs' quantities across any pair of sectors buying from the same sector(s) is driven by their ratio of intermediate intensities with whom the pair is purchasing from: $\mathcal{F} = \mathcal{D}^{fd}$.

LEMMA 2 (Factor input supply)

Changes in the ratio of intermediate input profits across any pair of sectors selling to the same sector(s) is driven by their ratio of intermediate intensities with whom the pair is trading to: $\mathcal{R} = \mathcal{D}^{fs}$.

► Go back, demand

► Go back, supply

- ▶ On the theoretical side, each cell identifies “network proximity”
 - the higher $d[s, s']$, the larger the transmission $\longleftrightarrow$ stronger horizontal complementarity
- ▶ Once taken to the data, each cell measures “network distance”, computed as:
 - Euclidean distance, $d^{fd}[s, s'] = \sqrt{[\alpha(s, s) - \alpha(s', s)]^2 + \dots + [\alpha(s, m) - \alpha(s', m)]^2}$;
 - each matrix is *symmetric* outside the main diagonal, with $\text{diag}(\mathcal{D}^j) = 0$.
- ▶ These are the *intensive margin*, with values inside $[0, \infty)$. To account for *extensive margin* (discrete) values, I employ the *shortest path algorithm*
 - that is, how many steps are necessary for a sector to reach another one
- ▶ Horizontal dimension is not weighted by vertical dimension (Theorem 3)

▶ Matrix configuration

▶ Theorem 2

- The *network demand-distance matrix* is defined as

$$\mathcal{D}_{m \times m}^{fd} = \begin{bmatrix} \left(\frac{\alpha(s,s)}{\alpha(s,s)}, \dots, \frac{\alpha(s,m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(s,s)}{\alpha(s',s)}, \dots, \frac{\alpha(s,m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(s,s)}{\alpha(m,s)}, \dots, \frac{\alpha(s,m)}{\alpha(m,m)} \right) \\ \left(\frac{\alpha(s',s)}{\alpha(s,s)}, \dots, \frac{\alpha(s',m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(s',s)}{\alpha(s',s)}, \dots, \frac{\alpha(s',m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(s',s)}{\alpha(m,s)}, \dots, \frac{\alpha(s',m)}{\alpha(m,m)} \right) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{\alpha(m,s)}{\alpha(s,s)}, \dots, \frac{\alpha(m,m)}{\alpha(s,m)} \right) & \left(\frac{\alpha(m,s)}{\alpha(s',s)}, \dots, \frac{\alpha(m,m)}{\alpha(s',m)} \right) & \cdots & \left(\frac{\alpha(m,s)}{\alpha(m,s)}, \dots, \frac{\alpha(m,m)}{\alpha(m,m)} \right) \end{bmatrix} = [d^{fd}(s, s')]$$

- Under due adjustments, analogous is the supply-distance matrix: $\mathcal{D}_{m \times m}^{fs} = [d^{fs}(s, s')]$

THEOREM 1 (On the dual representation)

Consider the equilibrium of an Input-Output economy, $\mathbf{y} = \mathbf{A}\mathbf{y} + \mathbf{c}$. The matrix $\mathbf{A}$ admits two distinct representations:

- vertical system, an equilibrium mapping governed by $\mathcal{L} = \sum_{k=0}^{\infty} \mathbf{A}^k$ encoding multiple production layers for circulating commodities;
- horizontal embedding, a collection of Input-Output profiles $\{\boldsymbol{\alpha}(s)\}_{s \in \Phi}$ encoding a geometric representation of sectoral similarities;

Both representations are fully determined by the primitive Input-Output matrix.

THEOREM 2 (On the configuration of network “economic” distances)

Let the set of sectors being Φ , and $\alpha^j(\cdot)$, for $j = \{fd, fs, \}$, the normalized Input-Output fixed coefficients. Then, demand- and supply-based distance matrices satisfy: (i) for any $s, s' \in \Phi$ there exists a common neighbour- s^* , so that both $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$ are non-negative and full; (ii) a given Input-Output structure uniquely determines $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$, which differ entrywise unless the corresponding bilateral trade with the common neighbour coincides; and (iii) if $\alpha^j(\cdot)$'s are non-symmetric, then all entries of $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$ differ; conversely, if $m \geq 4$, then both share at least one identical entry whenever at least $m - 2$ sectors trade symmetrically with a common sector and at least one additional symmetric trade relation elsewhere in the network exists.

COROLLARY 1 (Coinciding distances)

For $m \geq 4$, if a common sector trades symmetrically with $S = m - 2$ others, then $\frac{S(S-1)}{2}$ sectors exhibit symmetric distances.

- ▶ So far intermediate inputs generally function as substitutes
 - non-unitary substitution elasticity alters propagation {e.g., [Atalay \(2017\)](#), [Baqaei and Farhi \(2019\)](#)}
- ▶ Reduced-form intuition for opposite comovement under distances

► Propositions

$$\text{demand-driven comovement : } \underbrace{\frac{\partial \log x(s, s^*)}{\partial \log x(s', s^*)}}_{<0} \propto -\sigma [1 - e(s, s^*)] \underbrace{\frac{\partial \log p(s^*)}{\partial \log x(s', s^*)}}_{>0}$$

$$\text{supply-driven comovement : } \underbrace{\frac{\partial \log x(s^*, s)}{\partial \log x(s^*, s')}}_{<0} \propto \sigma e(s^*, s') \underbrace{\frac{\partial \log p(s')}{\partial \log x(s^*, s')}}_{<0}$$

👉 Substitution elasticities matter in magnitude, not in the direction of comovement

PROPOSITION 1 (Demand-driven comovement)

In the single-input case, comovement in demand implies $\delta_{s^ \rightarrow [s, s']} \propto -\sigma \psi^{fd}$. In the multi-input CES case, then $\delta_{s^* \rightarrow [s, s']} \propto -\sigma [1 - e(s, s^*)] \psi^{fd}$, with relative intermediate inputs changes from common supplier, $\delta_{s^* \rightarrow [s, s']} = \frac{\partial \log x(s, s^*)}{\partial \log x(s', s^*)}$, jointly determined by substitution elasticity σ , downstream sectoral expenditure share $e(\cdot)$, and upstream pass-through ψ^{fd} . Opposite comovement occurs if $\psi^{fd} > 0$.*

PROPOSITION 2 (Supply-driven comovement)

In the single-input case no condition exists for comovement in supply due to the absence of competition for downstream markets. In the multi-input CES case it holds $\delta_{\rightarrow s^} [s, s'] \propto \sigma e(s^*, s') \psi^{fs}$, with relative intermediate inputs changes to common buyer, $\delta_{\rightarrow s^*} [s, s'] = \frac{\partial \log x(s^*, s)}{\partial \log x(s^*, s')}$, jointly determined by substitution elasticity σ , sectoral revenues for upstream supplier $e(\cdot)$, and downstream pass-through ψ^{fs} . Opposite comovement occurs if $\psi^{fs} < 0$.*

- ▶ Intuition beyond the horizontal transmission:
 - *demand distance*: shocks trigger downstream buyer to shift away from upstream suppliers
 - ★ vertical and horizontal effects get mixed, making comovement ambiguous to interpret
 - *supply distance*: upstream shocks pass downstream, where inputs are used together
 - ★ horizontal propagation among suppliers is direct, mechanical, and unambiguous
- ▶ Core contrast:
 - *demand distance*: mere variation in upstream price (no true horizontal complementarities);
 - *supply distance*: substitution of upstream inputs through downstream demand reshuffling.
- ▶ Downstream complementarities (*i.e.*, supply distances) matter the most to generate opposite comovement {in line with [Atalay \(2017\)](#) and [Baqae and Farhi \(2019\)](#)}

► *Demand-based network distances*

- capture how easily a downstream buyer can switch among suppliers, without creating “true” horizontal linkages $\longleftrightarrow$ *upstream pass-through effect*;
- reflect substitution patterns across buyers sharing common upstream suppliers;
- ambiguous propagation effects, as they lack genuine complementarities among buyers.

► *Supply-based network distances*

- generate genuine horizontal complementarities among sectors whose outputs are jointly used by common downstream buyers $\longleftrightarrow$ *downstream pass-through effect*.
- shocks to one supplier propagate horizontally as other suppliers adjust production due to downstream complementarities;
- strong comovement effects, consistent with non-linear network theories emphasizing complementarities in intermediate inputs. {e.g., [Atalay \(2017\)](#), [Baqae and Farhi \(2019\)](#)}

- ▶ Equilibrium adjustments are central to understand the origins of aggregate cycles
- ▶ Fluctuations in aggregate real GDP in the model {e.g., vom Lehn and Winberry (2022)}:

$$d \log Y = \sum_{s \in \Phi} \left(\lambda(s) d \log z(s) + \nu(s) d \log n(s) \right)$$

that is, under the network's horizontal geometry,

$$d \log Y = \sum_{s \in \Phi} \lambda(s) d \log z(s) + \sum_{s \in \Phi} \begin{cases} \nu(s) \Delta \text{comovement} & \text{under major distance} \\ \nu(s) \Delta \text{reallocation} & \text{under minor distance} \end{cases}$$

👉 different network distance values generate different fluctuations in aggregate output

- ▶ How does sectoral employment fluctuate? $\longleftrightarrow$ *vertical* + *horizontal* dimensions

- ▶ Overall network effects on sectoral comovement of employment:

► Components

$$d \log \mathbf{n} = \Theta \left\{ d \log C + d \log \mathbf{S} + d \log \mathbf{V} + d \log \mathbf{D}(j) \right\}, \quad \text{for } j = \{fd, fs\}$$

with $\mathbf{D}(fd) = \mathbf{D}^{fd} \left[d \log \mathbf{n} - (\mathbf{I}'_{\Phi} d \log \mathbf{n}) \mathbf{1} \right]$ and $\mathbf{D}(fs) = \mathbf{D}^{fs} \left[d \log \mathbf{p} - (\mathbf{I}'_{\Phi} d \log \mathbf{p}) \mathbf{1} \right]$ denoting the difference between the change in all other sectors and the sector's own change

- ▶ Consistent interpretation:
 - *ambiguous* comovement in demand, as employment changes enter in both $\mathbf{V}$ and $\mathbf{D}(fd)$;
 - *opposite* comovement in supply, due to sectoral relative-price effects in $\mathbf{D}(fs)$;
 - vertical propagation prevails under low proximity, leading to *positive* comovement.

- (a) $\mathbf{n} = [n(s), n(s'), \dots, n(m)]$ identifies an $m \times 1$ vector of sectoral employment levels;
- (b) $\Theta = \frac{1}{1+\phi+\alpha}$ is a compound of structural parameters, and $\underset{m \times 1}{\boldsymbol{\alpha}} = [\alpha(s), \dots, \alpha(m)]$;
- (c) C is aggregate consumption;
- (d) $d \log \boldsymbol{\mathcal{S}} = d \log \mathbf{z} - d \log \mathbf{c}$ identifies the $m \times 1$ vectors for sector-specific changes in productivity and consumption, respectively;
- (e) $d \log \boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}}(\Phi_{\{s,s'\}}) [d \log \mathbf{z} + \underset{m \times 1}{\boldsymbol{\alpha}} d \log \mathbf{n}] + \boldsymbol{\mathcal{E}} d \log \mathbf{X}$ identifies the production network effect of other sectors' changes in productivities, employment levels, and intermediate inputs usage, where:
 - $\boldsymbol{\mathcal{A}}(\Phi_{\{s,s'\}})$ is the Input-Output matrix, comprising the weight of each sector on other sectors, whose entries are set to 0 whenever $s = s'$;
 - $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{A}}(\Phi_{\{s,s'\}})' \boldsymbol{\mathcal{A}}(\Phi_{\{s',s\}})$ is a compounded network effect, made of the inner product of the Input-Output matrix, influencing the elements of $\mathbf{X}$.

THEOREM 5: AGGREGATE FLUCTUATIONS [▶ GO BACK](#)

A

THEOREM 5 (On aggregate cycles)

Consider an Input-Output economy made of m -sectors. Let $\lambda(s)$ denote sector- s 's Domar weight, and assume no sector is dominant, in the sense that $\lambda(s) \asymp \frac{1}{m}$, $\forall s \in \Phi$. Beside vertical propagation, if sectors share upstream suppliers or downstream buyers, then idiosyncratic shocks generate non-vanishing aggregate GDP fluctuations: $\liminf_{m \rightarrow \infty} \text{var}(Y_m) > 0$. As many horizontally-related sectors move in response, shocks do not average out, and aggregate volatility survives even as the economy grows.

COROLLARY 3 (Granular sectoral weights)

Economically large sectors amplify aggregate fluctuations induced by horizontal network propagation.

COROLLARY 4 (Rate of decay)

Let Ψ be the set of horizontally connected sectors whose cardinality is $|\Psi|$. Horizontal geometry alone breaks the standard diversification argument: $\text{var}(Y_m) = \Omega(|\Psi|/m)$. Consequently, aggregate volatility decays at slower rate than $\mathcal{O}(1/m)$, the standard diversification rate.

THEOREMS 3 AND 4: HORIZONTAL WEIGHTING AND IMPORTANCE

A

THEOREM 3 (On the weighting of network “economic” distances)

In an Input-Output economy, the horizontal dimension of a network (defined by demand- and supply-based distance matrices) is not weighted by its vertical dimension (defined by the directed or Leontief inverse matrix).

THEOREM 4 (On the importance of network “economic” distances)

Consider a network of size- m with Input-Output matrix $\mathbf{A}_m := \mathbf{V}_m \cup \mathbf{D}_m$, where $\mathbf{V}_m$ encodes vertical buyer-supplier ties and $\mathbf{D}_m$ encoding horizontal complementarities. Let $\boldsymbol{\varepsilon}_m$ be a vector of independent shocks, and let $\mathcal{P}_m^{(\mathcal{V})}$ and $\mathcal{P}_m^{(\mathcal{D})}$ denote the associated matrices for propagation channels. Then, as the network size grows, the vertical propagation vanishes in spectral norm:

$$\left\| \mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\varepsilon}_m \right\|_2 / \left\| \mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\varepsilon}_m \right\|_2 \xrightarrow{m \rightarrow \infty} 0.$$

COROLLARY 2 (Endogenous expansion)

In growing networks, the horizontal geometry asymptotically dominates shock propagation even as vertical chains expand.

► market share, after redefinitions Input-Output tables

- square matrix, $\mathbf{X}_{m \times m}$, whose diagonal correspond to sector-share of own production while:
 - ★ $\mathbf{X}_{m \times m}^{up} = rows(\mathbf{X})$ is the share of production of sector-s that is bought by sector-s';
 - ★ $\mathbf{X}_{m \times m}^{dw} = columns(\mathbf{X})$ is the share of production that sector-s buys from sector-s'.
- in the dataset built, I compute $\mathbf{x} = \mathbf{1} - diag(\mathbf{X})$ to extract the *total share* of production that sectors buy from other sectors, where $\mathbf{1}$ an $m \times 1$ vector of ones

► commodity-by-commodity total direct requirements Input-Output tables

- square matrix, $\mathbf{A}_{m \times m}$, in which each entry corresponds to:
 - ★ $\mathbf{A}_{m \times m}^{up} = rows(\mathbf{A})$ is the dollar value of one unit of sector-s that is bought by sector-s';
 - ★ $\mathbf{A}_{m \times m}^{dw} = columns(\mathbf{A})$ is the dollar value of sector-s in order to buy one unit from sector-s'

→ *row sectors identify supplier (upstream), and column sectors identify buyer (downstream)*

INTERPRETING AN INPUT-OUTPUT MATRIX, CELLS

B - 2/3

	AGRICULTURE	MANUFACTURING	SERVICES
AGRICULTURE		AGRICULTURE SELLS TO MANUFACTURING	AGRICULTURE SELLS TO SERVICES
MANUFACTURING	AGRICULTURE BUYS FROM MANUFACTURING		MANUFACTURING SELLS TO SERVICES
SERVICES	AGRICULTURE BUYS FROM SERVICES	MANUFACTURING BUYS FROM SERVICES	

► Go back

- ▶ To investigate the network structure:
 - upstream and downstream sectors, as well as sectoral distances, are computed through the *commodity-by-commodity direct requirements* table as of 2007;
 - only to compute the Leontief inverse, disregard small transactions

$$\alpha^{new}(s, s') = \frac{\alpha(s, s')}{\alpha(s)} \text{ is set to } 0 \text{ if } \alpha^{new}(s, s') \leq 0.01$$

where $\alpha(s) = \sum_{s' \in row} \alpha(s, s')$ or $\alpha(s) = \sum_{s' \in column} \alpha(s, s')$

- ▶ Defining $\boldsymbol{\alpha} = [\alpha(s), \alpha(s'), \dots, \alpha(m)]'$, the considered Input-Output structure is

$$\underset{m \times m}{\mathcal{A}}^{new} = \underset{m \times m}{\mathcal{A}}' \underset{m \times 1}{\boldsymbol{\alpha}}^{-1}$$

in which each cell, $\alpha^{new}(s, s')$, is greater than 1%

share of sectoral total sales

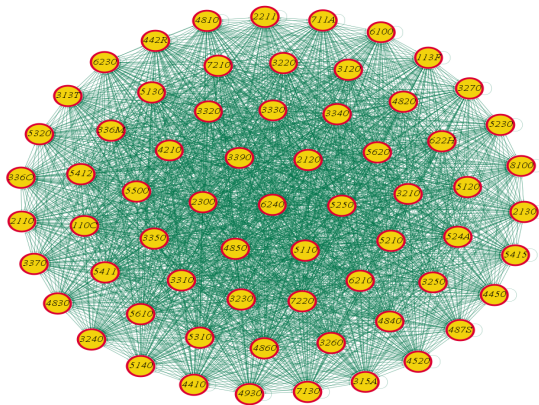


Figure B.1: Production network of the U.S. economy in 2007

► [Go back](#)

NETWORK REPRESENTATION

share of sectoral total purchases

B

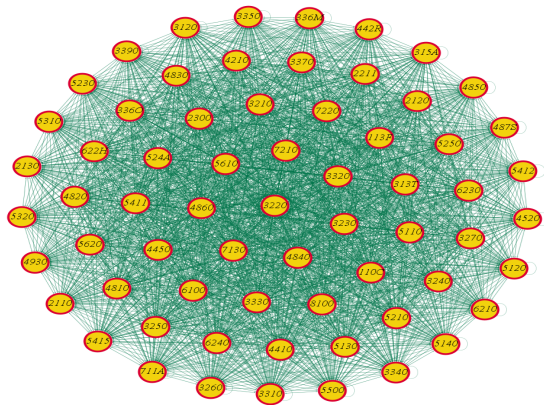


Figure B.2: Production network of the U.S. economy in 2007

► Go back

- System of *Cartesian coordinates* to compute the *network distances* {e.g., [Conley and Dupor \(2003\)](#)}:

- ① *factor input demand*, when buying similar inputs. Given $\alpha^{fd}(s, s') = \frac{\alpha(s, s')}{\sum_{s' \in \text{row}} \alpha(s, s')}$, then

$$d^{fd}[s, s'] = \left\{ \sum_{s^* \in \Phi} \left[\alpha^{fd}(s, s^*) - \alpha^{fd}(s', s^*) \right]^2 \right\}^{\frac{1}{2}}$$

is measuring downstream proximity of production function requirements.

► Heatmap, demand

- ② *factor input supply*, when selling to similar sectors. Given $\alpha^{fs}(s, s') = \frac{\alpha(s, s')}{\sum_{s' \in \text{column}} \alpha(s, s')}$, then

$$d^{fs}[s, s'] = \left\{ \sum_{s^* \in \Phi} \left[\alpha^{fs}(s^*, s) - \alpha^{fs}(s^*, s') \right]^2 \right\}^{\frac{1}{2}}$$

is measuring upstream comovement of sectoral revenues from trade;

► Heatmap, supply

- Distances are computed as those from the *directed network*

► Distributions

► Correlation

- the Leontief inverse transformation, instead, amplifies the magnitude of interconnections

► Go back

SECTORAL DISTANCE IN THE NETWORK, FACTOR INPUT DEMAND

B - 1/2

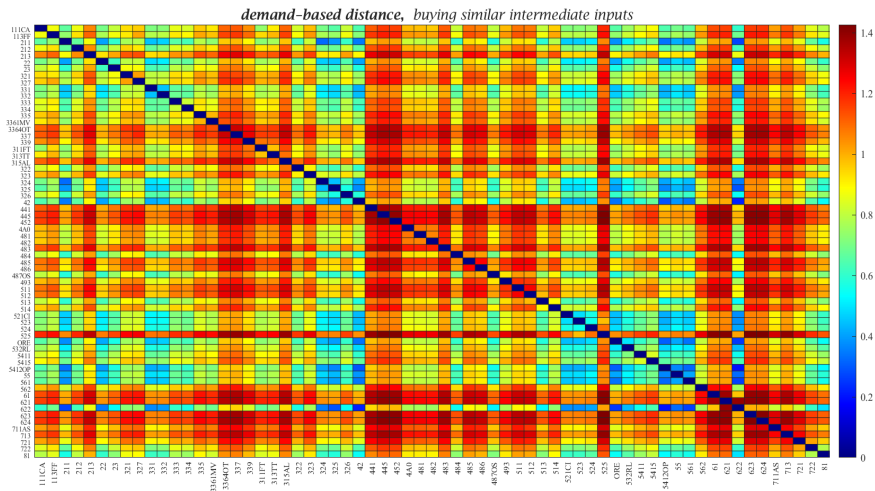


Figure B.3: Factor demand distances in the Input-Output matrix, U.S. 2007

SECTORAL DISTANCE IN THE NETWORK, FACTOR INPUT SUPPLY

B - 2/2

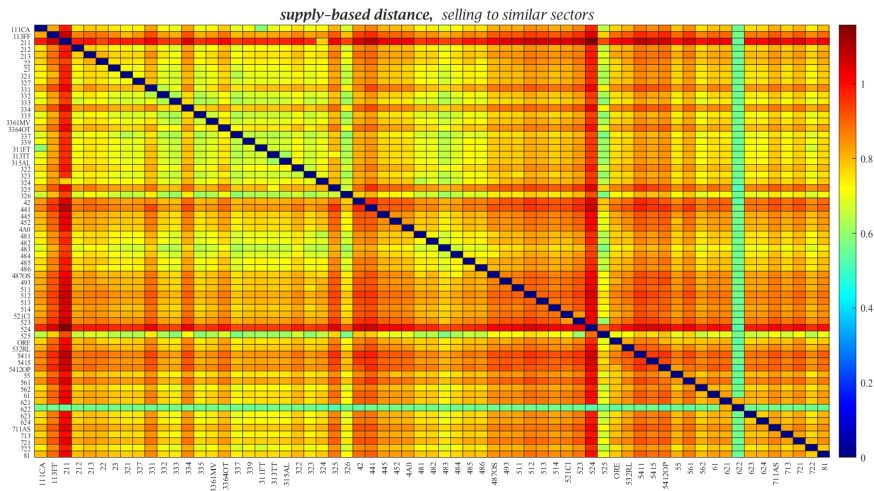
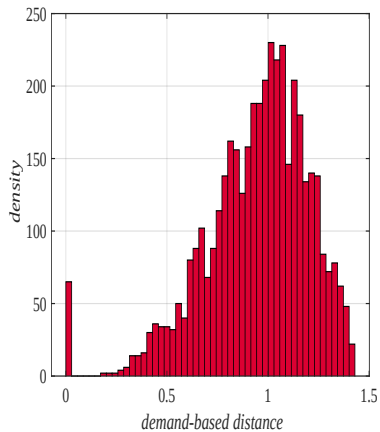
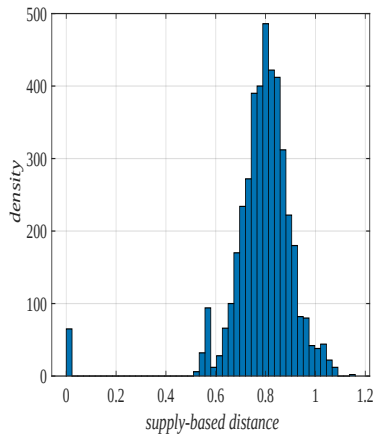


Figure B.4: Factor supply distances in the Input-Output matrix, U.S. 2007



(a) Factor input demand



(b) Factor input supply

Figure B.5: Distribution of network economic distances

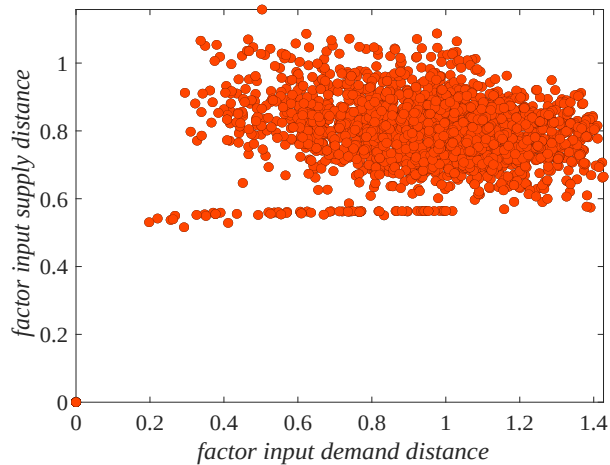


Figure B.6: Correlation of network economic distances

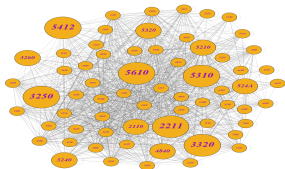
Shortest path algorithm.

Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other node in a network. In particular:

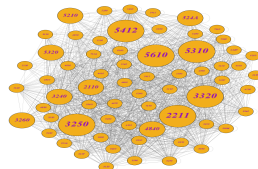
- (a) identify the direct connections between any pair of nodes (those that take exactly one step to be connected);
- (b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.).

Multiply the graph by itself to discover which new pairs of nodes are now connected through longer paths;

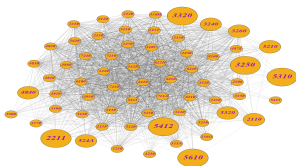
- (c) if a pair of nodes becomes connected (*i.e.*, only if it has not be already found a shorter path), record the current length as the shortest distance between them;
- (d) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still unreachable (*i.e.*, the distance is still zero and they are not linked in the network), set their distance to infinity.



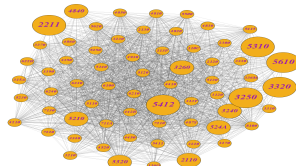
(a) Demand, $d = 1$



(b) Supply, $d = 1$



(c) Demand, $d = 2$



(d) Supply, $d = 2$

Figure B.7: Visualization of production network distances

- (1) The distribution of horizontal complementarities is uneven
 - the factor input demand network is slightly denser than the supply network at $d = 1$, but then this asymmetry vanishes ($d = 2$)
 - (2) Sectors with the major Input-Output linkages tend to concentrate around the nucleus
 - the most connected hubs (top 10%) and the mid-hubs (top 20%) are similarly central
 - (3) Sectors with the highest number of production network connections tend to be evenly central at shorter distances, and relatively more peripheral at longer distances
- 👉 The structure of horizontal complementarities depend on the notion of distance
- the same vertical production system generates distinct horizontal geometries (Theorem 2)

- ▶ Vertical propagation amplify shocks through supply chains and cascade effects, while horizontal geometry alone prevents sectoral fluctuations from vanishing in aggregate
- ▶ Granularity mechanically increases aggregate volatility
 - however, horizontal spillovers are mitigated when large Domar weights are associated with sectors embedded in large horizontal clusters (the “*h*-cluster”)
- ▶ Horizontal propagation generates larger aggregate responses
 - vertical transmission amplifies shocks through recursive cascades but produces “localized” effects along specific supply chains;
 - horizontal propagation increases systemic covariance, creating widespread comovement without confined feedback loops or cascading amplification.
- ▶ $\|\mathcal{P}^{(\mathcal{D})}\|_2 = 0.156 > 0.124 = \|\mathcal{P}^{(Lucas, 1977)}\|_2$
 - horizontal propagation induce aggregate fluctuations to mean revert at a slower rate of decay than predicted by the standard diversification argument

Table B.1: Dimensional dominance

	<i>network sparsity</i>				
	15%	30%	40%	50%	95%
	$\tau = 0.5\%$	$\tau = 1\%$	$\tau = 1.5\%$	$\tau = 2\%$	$\tau = 5\%$
<i>vertical propagation</i>	.2133	.1995	.1803	.1497	.0762
<i>horizontal propagation</i>	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007 when small transactions, at different threshold- τ , are removed. Source: BEA and own calculations.

① *Aggregate TFP shock*

- *supply-based* horizontal exposure: large, positive, and *persistent*
 - ★ sectors sharing common downstream buyers react more strongly to TFP innovations
- *demand-based* horizontal exposure: statistically insignificant at all horizons
- *vertical* (Leontief inverse) exposure: positive but smaller, quickly dissipating
- 👉 aggregate productivity diffuses primarily through *common downstream demand* linkages, not through observed Input-Output chains

② *Government spending shock*

- *demand-based* horizontal exposure: positive, significant, and *persistent*
 - ★ sectors with common suppliers to government-exposed industries amplify fiscal shocks
- *supply-based* horizontal exposure: negative, significant, and *persistent*
 - ★ weaker responses, consistent with competitive demand reallocation
- *vertical* (Leontief inverse) exposure: statistically insignificant at all horizons
- 👉 horizontal geometries are the *primary* propagation channel for fiscal disturbances; dominant channel varies across shock types

- ▶ sectoral output and employment growth, $\dot{var}_t(s) = var_t(s) - var_{t-1}(s)$, $var = \{y, n\}$;
- ▶ network distance- d computed as a *shortest path problem*, unweighted (Theorem 3);
- ▶ given Φ_s^j the set of sectors which are $k \in \{upstream, downstream\}$ to sector- s :
 - (a) distance- d growth, $\dot{var}_t(\Phi_s, d) = \sum_{s' \neq s} [var_t(s', d) - var_{t-1}(s', d)]$;
 - (b) type- k growth, $\dot{var}_t(\Phi_s, k) = \sum_{s' \neq s} [\ell(s, s' | k) (var_t(s', k) - var_{t-1}(s', k))]$;

with $\ell(s, s')$ being the Leontief inverse weight of sector- s' on sector- s
- ▶ controls:
 - (a) sectoral employment size, $n(s)$, value-added $y(s)$, and their growth rates, and net exports;
 - (b) aggregate employment size N and value-added Y , and their growth rates.

① Purge the dynamics of sectoral I-O shares from movements in sectoral employment

- panel Fixed-Effects (FE) regression over $s \in \Phi$ sectors

$$\begin{aligned}
 x_t(s) = & \underbrace{\beta_n \dot{n}_t(s)}_{\text{emp. growth}} + \underbrace{\sum_d \beta_{n(d)} \dot{n}_t(\Phi_s, d)}_{\text{emp. growth, network distance}} + \underbrace{\sum_k \beta_{n(k)} \dot{n}_t(\Phi_s, k)}_{\text{emp. growth, upstream and downstream}} \\
 & + \underbrace{\beta_{z(s)} \mathcal{Z}_t(s) + \beta_{\dot{z}(s)} \dot{\mathcal{Z}}_t(s)}_{\text{sector-level controls}} + \underbrace{\beta_z \mathcal{Z}_t + \beta_{\dot{z}} \dot{\mathcal{Z}}_t}_{\text{aggregate controls}} + \underbrace{\phi(s)}_{\text{sector Fe}} + u_t^x(s)
 \end{aligned}$$

- estimated residuals $\left(\hat{u}_t^x(s)\right)$ identify the residual variation in I-O shares

- ② Estimate panel Local Projections (LP) to study the dynamic effects of changes in sectoral I-O intermediate shares on its (cumulative) employment
- given $h = 5$ horizons, sequence of panel predictive regressions of the form

$$\dot{n}_{t+h}^{cum}(s) = \beta_{\hat{u}^x, h} \hat{u}_t^x(s) + \underbrace{\phi(t+h) + \phi_h(s)}_{\text{horizon and sector Fe}} + v_{t+h}^x(s)$$

... also considering d^{th} -distance across sectors

$$\dot{n}_{t+h}^{cum}(s) = \beta_{\hat{u}^x(d), h} \sum_{s' \neq s} \hat{u}_t^x(s', d) + \underbrace{\phi(t+h) + \phi_h(s)}_{\text{horizon and sector Fe}} + v_{t+h}^x(s, d)$$

- variables at h identifies the cumulative difference between time t and $t + h$

NETWORK SHARES AND EMPLOYMENT

factor input demand and directed network

► OVERVIEW

C - 1/2

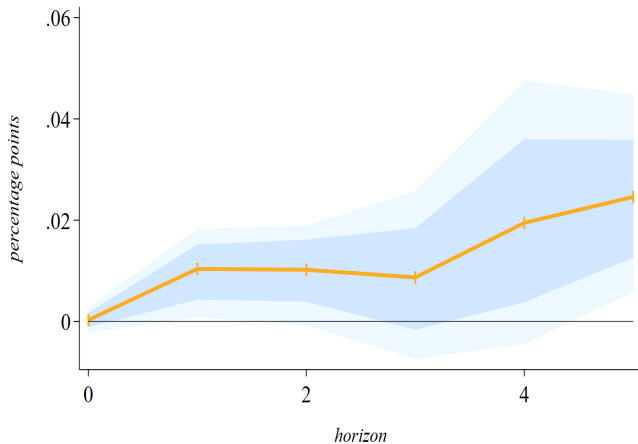


Figure D.1: Sectoral employment response following a 1% increase in own I-O share

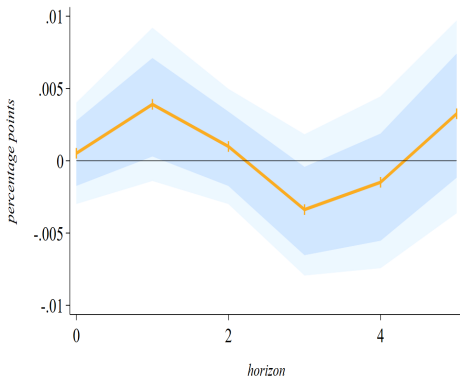
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NETWORK SHARES AND EMPLOYMENT, DISTANCE

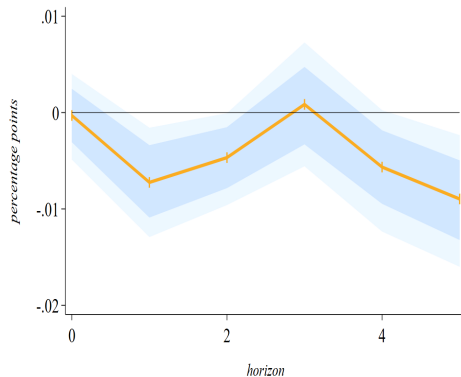
factor input demand and directed network

► OVERVIEW

C - 2/2



(a) $d = 1$



(b) $d = 2$

Figure D.2: Sectoral employment response following a 1% increase in I-O share of nearby sectors

NETWORK SHARES AND EMPLOYMENT

factor input supply and directed network

► OVERVIEW

C - 1/2

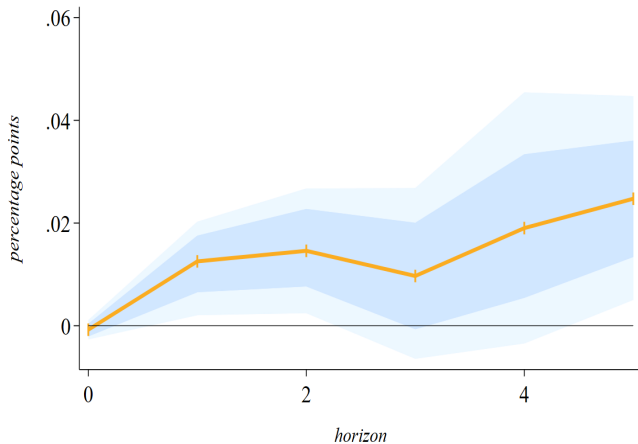


Figure D.3: Sectoral employment response following a 1% increase in own I-O share

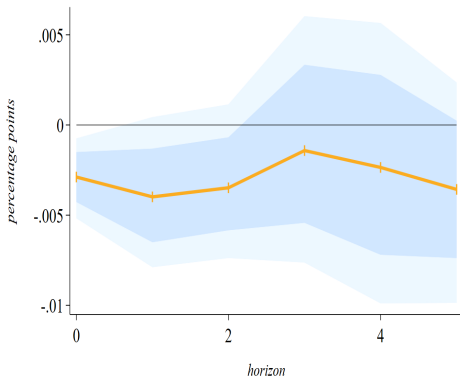
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NETWORK SHARES AND EMPLOYMENT, DISTANCE

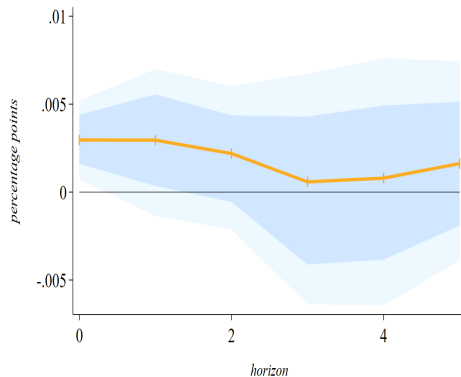
factor input supply and directed network

► OVERVIEW

C - 2/2



(a) $d = 1$



(b) $d = 2$

Figure D.4: Sectoral employment response following a 1% increase in I-O share of nearby sectors

- ① IDENTIFICATION, to extract exogenous dynamics {as in Barattieri and Cacciatore (2023)}

$$\begin{aligned}
 n_t(s) = & \underbrace{\beta_y \dot{y}_t(s)}_{\text{output growth}} + \underbrace{\sum_d \beta_{y(d)} \dot{y}_t(\Phi_s, d)}_{\text{output growth, network distance}} + \underbrace{\sum_k \beta_{y(j)} \dot{y}_t(\Phi_s, k)}_{\text{output growth, upstream and downstream}} \\
 & + \underbrace{\beta_{z(s)} \mathcal{Z}_t(s) + \beta_{\dot{z}(s)} \dot{\mathcal{Z}}_t(s)}_{\text{sector-level controls}} + \underbrace{\beta_z \mathcal{Z}_t + \beta_{\dot{z}} \dot{\mathcal{Z}}_t}_{\text{aggregate controls}} + \underbrace{\phi(s)}_{\text{sector Fe}} + u_t^n(s)
 \end{aligned}$$

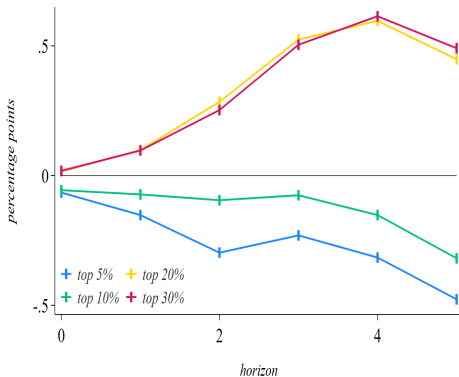
- estimated residuals $\left(\hat{u}_t^n(s)\right)$ identify the residual variation in sectoral employment

- ② LOCAL PROJECTIONS, for short-run dynamic responses {as in Barattieri and Cacciatore (2023)}

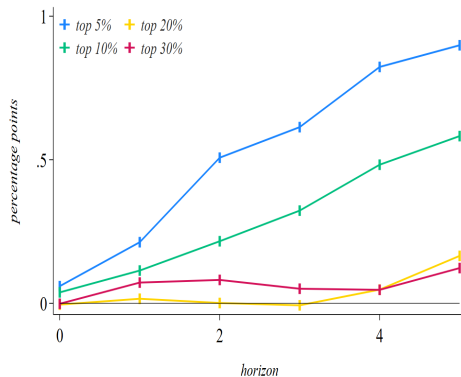
$$\dot{n}_{t+h}^{cum}(s) = \beta_{\hat{n}(d),h} \sum_{s' \neq s} \hat{u}_t^n(s', d) + \underbrace{\phi(t+h) + \phi_h(s)}_{\text{horizon and sector Fe}} + v_{t+h}^n(s, d)$$

EMPLOYMENT EFFECTS OF MOST CONNECTED SECTORS, DISTANCE

factor input demand and directed network



(a) $d = 1$, directed

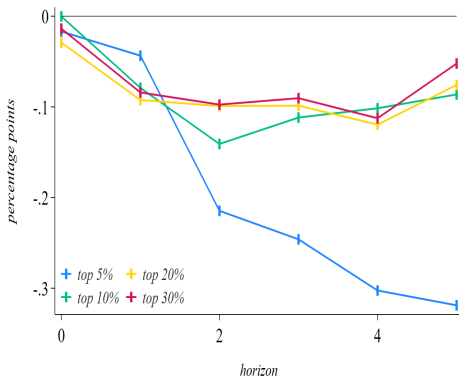


(b) $d = 2$, directed

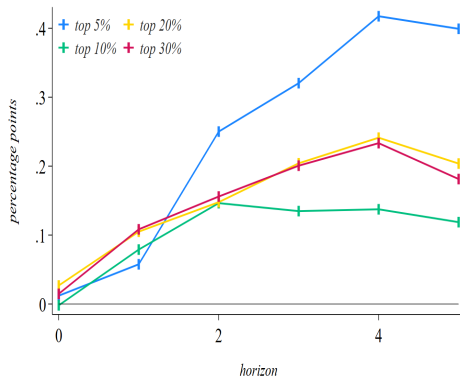
Figure E.1: Demand linkages and comovement in interlinked sectors

EMPLOYMENT EFFECTS OF MOST CONNECTED SECTORS, DISTANCE

factor input supply and directed network



(a) $d = 1$, directed



(b) $d = 2$, directed

Figure E.2: Supply linkages and comovement in interlinked sectors

► *peripheral sectors*

► Peripheral sectors

- *intuition*: a shock in a central sector transmits outward to sectors “at the margins” of the network, inducing broader fluctuations;
 - ★ if network distances were irrelevant, comovement would be expected universally;
- *results*: opposite comovement under both minor demand and supply distances; positive for shocks originated in further sectors.

► *inventories*

► Changes in private inventories

- *intuition*: inter-temporal buffer that smooths production responses, so that the contemporaneous nature of horizontal complementarities is attenuated
- *results*: horizontal transmission vanishes when holding inventories; in contexts without inventories, the baseline results are preserved

► *different base year*

in the paper

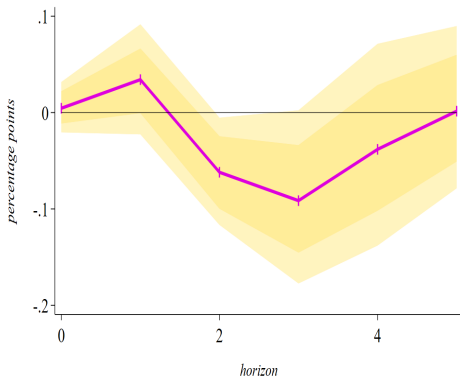
- *intuition*: the production network structure exhibits limited evolution over time;
- *results*: robust if replicated for alternative benchmark years (2002 and 2012).

EMPLOYMENT EFFECTS, DISTANCE

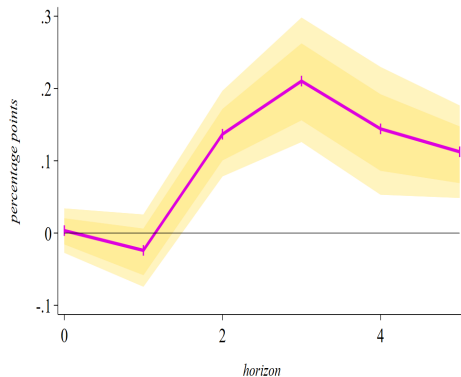
factor input demand and directed network

► OVERVIEW

C



(a) $d = 1$



(b) $d = 2$

Figure E.3: Employment comovement under direct demand linkages

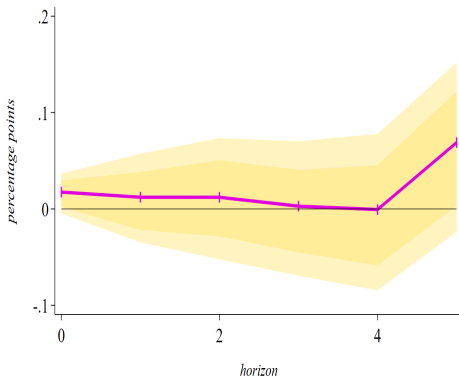
► Go back

EMPLOYMENT EFFECTS, DISTANCE

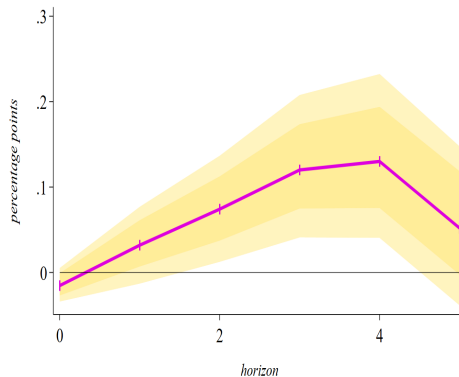
factor input supply and directed network

► OVERVIEW

C



(a) $d = 1$



(b) $d = 2$

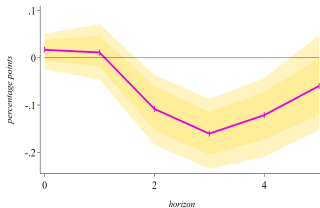
Figure E.4: Employment comovement under direct supply linkages

► Go back

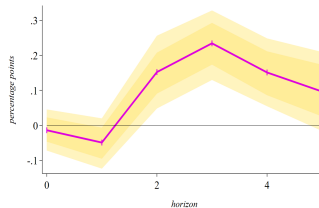
EMPLOYMENT EFFECTS IN MOST PERIPHERAL SECTORS, DISTANCE

factor input demand, factor input supply and directed network

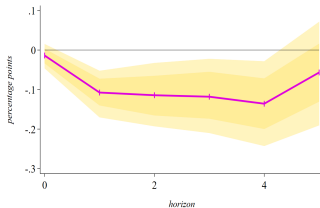
C



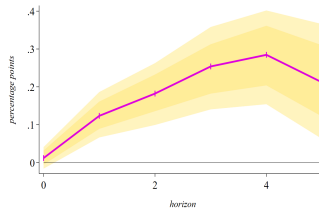
(a) $d = 1$, demand



(b) $d = 2$, demand



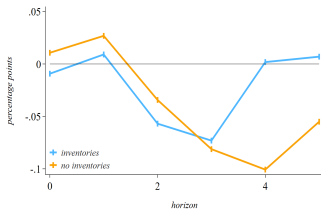
(c) $d = 1$, supply



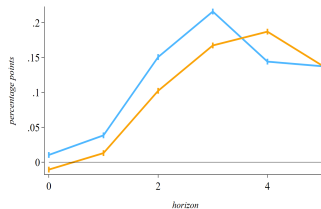
(d) $d = 2$, supply

EMPLOYMENT EFFECTS GIVEN CHANGES IN PRIVATE INVENTORIES, DISTANCE c

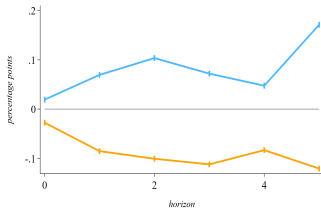
factor input demand, factor input supply and directed network



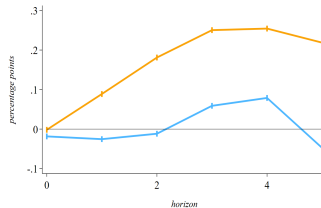
(a) $d = 1$, demand



(b) $d = 2$, demand



(c) $d = 1$, supply



(d) $d = 2$, supply